Utilizing Dual Active Bridge Converter for Electric Vehicle DC Charging Application

A PROJECT REPORT

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by

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VIT CHENNAI

APRIL 2025

CERTIFICATE

This is to certify that the project work titled **"Utilizing Dual Active Bridge Converter for EV DC Charging Applications"** submitted by **SRIDUTT LANKA, SUYASH PRAKASH** is in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY**, is a record of bona fide work done under my guidance. The contents of this project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for award of any degree or diploma and the same is certified.

Project Supervisor

The thesis is satisfactory / unsatisfactory

Internal Examiner **External Examiner**

Approved by

Head of the Department
B.Tech Electrical and Electronics Engineering

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I am appreciative of VIT Chennai and **Dr. Senthil Kumar N** [Dean/SELECT] for giving me chance to do this project, which has been a worthwhile educational experience.

We extend our heartfelt gratitude to all individuals who played a role in the success of our college project. Your invaluable support guidance have been greatly appreciated, and the completion of this project would not have been possible without your assistance. Thank you.

(Signature of the student)

Name of the Student

(Reg. Number)

ABSTRACT

The popularity of electric vehicles (EVs) as a sustainable mode of transportation is on the rise; however, their extensive adoption is hindered by factors such as the availability and efficiency of charging infrastructure. The dual active bridge (DAB) converter presents a promising solution for EV charging, as it is capable of functioning at elevated power levels, boasts high power density, and provides excellent efficiency, along with its bidirectional capabilities. This report outlines a vehicle-to-vehicle (V2V) charging mechanism for electric vehicles (EVs). The V2V charging technology offers a viable solution to the problem of inadequate charging station availability. It also enables charging at any time and in any location, much like a power bank. This system allows a four-wheeled EV battery to charge a two-wheeled EV battery, promoting bi-directional charging functionality.

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ABBREVIATIONS AND NOMENCLATURE

EV	Electric Vehicles
DAB	Dual Active Bridge
V2V	Vehicle to Vehicle
PV	Photovoltaic
UPS	Uninterruptible Power Supplies
EVSE	Electric Vehicle Supply Equipment
CV	Constant Voltage
SOC	State of Charge
ZVS	Zero Voltage Switching

CHAPTER I

1. INTRODUCTION

1.1 INTRODUCTION

The increasing demand for electricity, coupled with environmental issues and inconsistencies in energy production [1], has led to a growing interest in modern hybrid technologies for the conversion of renewable energy sources. A bidirectional DC/DC converter serves as a crucial interface between various renewable energy forms, such as fuel cells and photovoltaic (PV) arrays [2], and different energy storage systems, including batteries and supercapacitors [3]. These converters find applications in numerous fields, including hybrid vehicles, distribution networks, electric aircraft, and uninterruptible power supplies (UPS)[4]. The heightened need for intermediate electrical energy storage in battery systems, particularly driven by the integration of renewable energy [5], has created a demand for bidirectional DC/DC power converters that feature galvanic isolation [6]. In this context, a dual active bridge converter has been selected, allowing for power transfer in both directions [7]. This converter offers numerous advantages over alternative designs, including enhanced efficiency, circuit isolation, high-frequency operation, reliability, reduced component count, soft-switching capabilities, cost-effectiveness, and high performance [8]. This topology is particularly recommended for applications where power density, cost, weight, and reliability are of paramount importance [9].

1.1.1 Motivation

The design of a Dual Active Bridge (DAB) converter addresses critical challenges in cities within developing countries, such as India, where electric vehicle (EV) infrastructure remains underdeveloped. The absence of sufficient charging points often hinders the widespread adoption of EVs, especially two-wheelers, which are a predominant mode of transport in urban areas. By utilizing DAB converters, multiple DC voltage sources, such as batteries and Electric Vehicle Supply Equipment (EVSE), can be efficiently harnessed to facilitate EV charging. This converter enables seamless power exchange between various sources, providing flexibility and scalability for charging systems in areas with limited infrastructure. The result is an effective solution for optimizing energy distribution, improving EV accessibility, and supporting the growing need for reliable, adaptable charging systems in densely populated cities.

Additionally, the DAB converter offers transformative opportunities to redefine home and apartment charging setups. Instead of requiring individual EVSE installations, this converter enables the utilization of car batteries to charge smaller vehicle batteries, such as those in two-wheelers. By extending its capability to accommodate 400V batteries and solar-powered grids, the converter not only increases the number of charging nodes but also reinforces the use of renewable energy sources. This reduces dependency on extensive infrastructure while encouraging sustainable energy practices. The DAB converter thus emerges as an indispensable tool in the pursuit of efficient, cost-effective, and environmentally-conscious EV charging solutions, fostering progress in urban mobility and renewable energy adoption.

1.1.2 Objectives

- To design and simulate dual active bridge for DC charging purposes.
- To incorporate charging using a four-wheeler ev battery to charge a two-wheeler ev battery and facilitate bi-directional charging.
- Achieve CV charging via closed loop feedback.
- Replace Input Voltage source with 4-Wheeler Battery (Ideal) and output load with 2-Wheeler Battery (Ideal)

1.1.3 Scope of the Work

Bidirectional Power Transfer: The DAB converter has been meticulously designed to facilitate bidirectional power transfer between 400V and 60V sources. This design ensures compatibility for applications like battery charging and discharging scenarios, maintaining flexibility in energy flow.

Voltage Regulation and Battery Safety:

- The converter's control mechanism is optimized to maintain the voltages at rated levels, ensuring efficient battery charging.
- The input power is carefully managed to ensure that the voltage drop, especially at high states of charge (SoC), stays within acceptable thresholds. This prevents the voltage from exceeding the battery's safety limits, safeguarding battery longevity.
- Maintaining nominal voltages reduces the charging speed slightly, which, while deliberate, avoids damage caused by exposing the battery to high terminal voltages.

Mitigation of Wear and Losses:

- High input currents are avoided during charging to minimize wear and tear on the battery. Excessive current can lead to significant heat loss, resulting from increased (I^2R) losses due to the internal resistance of the battery.
- By regulating the charging current, thermal stress and degradation over time are reduced, promoting overall efficiency and battery lifespan.

Size Constraints and Trade-Offs:

- The converter design incorporates size-constrained components, which impacts its functionality and performance. Notably, the size constraint imposes a trade-off between the resolution of phase shift control and the power delivery capability.
- Larger values of leakage inductance in the design enable a finer resolution of phase shift, enhancing precision in power modulation. However, this comes at the cost of increased size, which can be a limiting factor in compact applications.

This balanced design ensures optimal performance while adhering to the safety and size constraints, delivering an efficient and sustainable solution for energy transfer and battery management.

1.2 Organization of Thesis

This thesis has been structured to present the project in a clear and systematic manner across the following chapters:

1. **Chapter 1: Converter and Control System**
   This chapter explains the key components of the project, focusing on the design and functionality of the DAB converter, along with its control system. Special emphasis is given to the utilization of a PI controller for regulating the current through the leakage inductance and the simplified transfer function employed for ease of understanding.

2. **Chapter 2: Converter Design and Trade-Offs**
   The design principles of the DAB converter are explored in detail in this chapter. Additionally, it discusses the trade-offs involved in the design, such as balancing the resolution of phase shifts and the size constraints of the converter.

3. **Chapter 3: Simulation Results**
   This chapter projects and analyzes the simulation results obtained for the DAB converter and control system, validating the design and operational efficiency of the project.

4. **Chapter 4: Summary and Future Scope**
   The concluding chapter summarizes the work carried out in the project and highlights the results achieved. It also provides insights into potential future improvements and extensions of the current work.

CHAPTER II

2. PROJECT DESCRIPTION

2.1 OVERVIEW OF PROJECT

- The charger under consideration incorporates a bi-directional mechanism aimed at efficiently charging the batteries of electric vehicles.

Figure 1- Block diagram of Dual Active Bridge converter

Figure 1 displays the flow of the Dual Active Bridge (DAB) converter, comprising of 3 blocks namely the primary bridge, the isolated transformer and the secondary bridge with 2 other blocks being the source and the load.

- A DC source is given as an input to the dual active bridge converter.
- By utilizing a full bridge converter, the direct current (DC) voltage is converted into a quasi-square-wave alternating current (AC), thereby increasing efficiency compared to hard-switched designs.
- The square wave voltage is then directed through a high-frequency transformer, a component crucial for ensuring galvanic isolation.
- The presence of leakage inductance in the transformer contributes to effective power transfer in both directions and helps to achieve soft switching.
- The square wave generated on the secondary side is subsequently transformed into direct current (DC) voltage through a full bridge converter, and the resulting voltage is then provided to the battery.

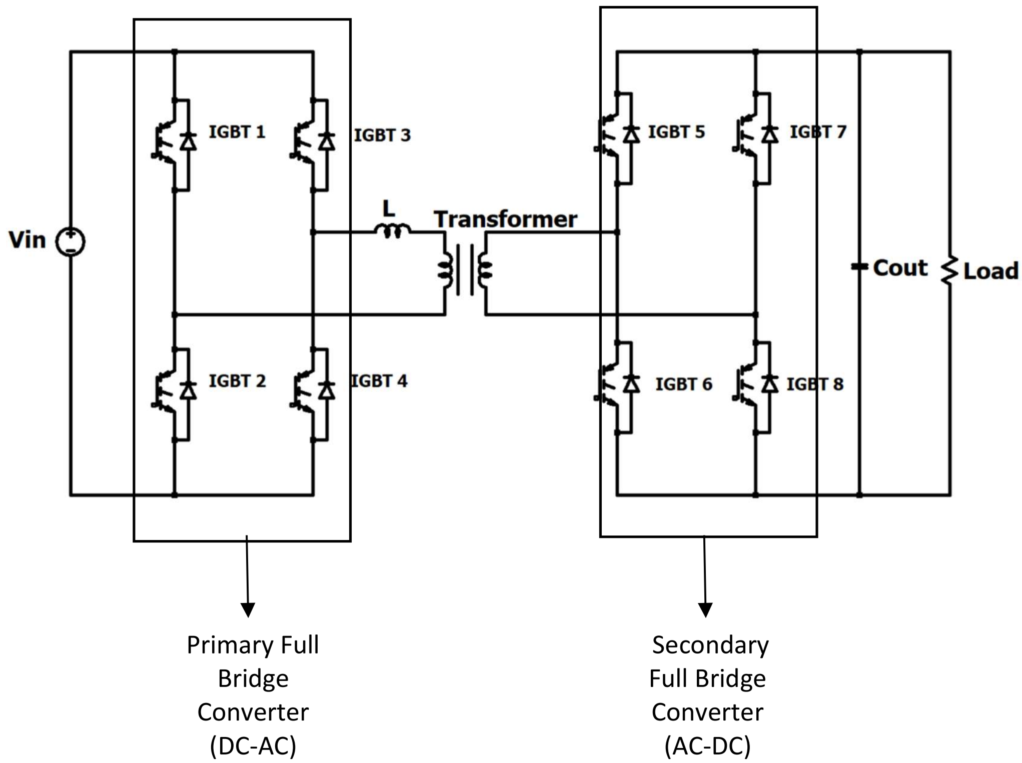


Figure 2- Circuit diagram for Dual Active Bridge Converter

Figure 2 displays the Dual Active Bridge (DAB) configuration being used in the project. The leakage inductance of the transformer is attached to the second arm of the primary bridge and the secondary coil is attached to the primary arm of the secondary bridge.

- The Dual Active Bridge is composed of two full bridges, referred to as the primary and secondary bridges. Each of these bridges contains four switches that operate using single phase shift modulation.
- This approach is easy to execute and provides excellent control performance. In circuits that employ this modulation technique, the square voltages will consistently have a duty cycle of 50% of the switching period, while the frequency is kept constant.

- The two bridges are managed with a defined phase difference, resulting in a voltage difference that allows current to move from the primary side to the secondary side.

2.2 Modules of the Project

2.1.1 Module 1

Module 1 consists of the converter and its transfer function which is derived from the leakage inductance current and the duty ratio of the phase shift that is explained from quarter of the switching cycle as D is limited to 0.5 to -0.5 considering the linear operational range of the converter.

The varying of the voltages at the output and input is assumed to be very negligible to the current control as the current control is faster than the variation and ripples at the voltage ports. This is due to the support at the voltage ports due to capacitors. This eases the modelling of the plant [10]. However, this model, although convenient, has some flaws due to its inaccuracies and not accounting for high-frequency components of the inductance current. This inevitably causes error; however, the control limits the power output to a suitable rating mitigating damage to the plant.

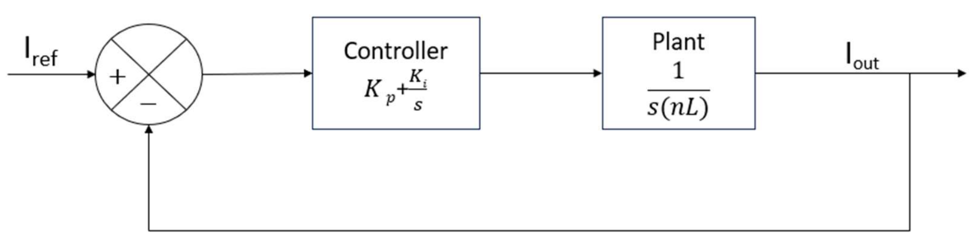


Figure 3- Closed Negative Feedback System of Plant and Controller

Figure 3 illustrates the negative feedback system employed in the control system where the output current is utilized to calculate the required phase difference between the primary and secondary bridge.

2.1.2 Module 2

Module 2 consists of the design of the control of the plant utilizing a PI controller. The basic plant model causes a noticeable error in the plant. To circumvent this, a feedforward method is further supplemented to help the control of the plant. The control switches from the PI loop to the feedforward control based on the reference and output signals. This compensates for the high settling time and inaccuracies of the plant model. A feedforward compensation technique also is satisfactory due to the DAB's inherent critically damped response upon simulation of open loop simulation. This shows a stark comprehension from the model thus paving the way to more robust stability of control without depending heavily on the approximate model of the plant.

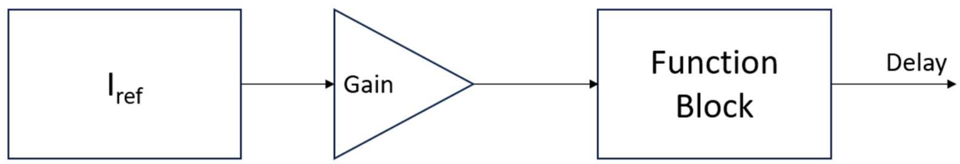


Figure 4- Feed-forward compensation technique

Figure 4 displays the feedforward technique being implemented in the control logic to allow quicker response keeping in mind the intrinsic critically damped response of the DAB converter.

The final closed-loop system can be represented by the plant as follows:

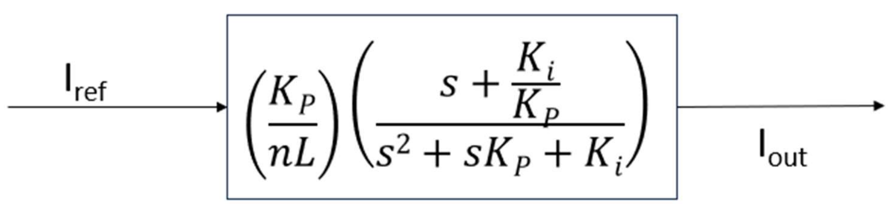


Figure 5- Closed Loop System Transfer Function

Figure 5 illustrates the entire closed loop feedback system of the plant. Utilising the required design specifications of the transient with the inherent critical damping ratio, the values of Kp and Ki can be calculated using the above transfer function.

CHAPTER III

3. Design of the Dual Active Bridge Converter

3.1 DESIGN APPROACH

The following design equations were implemented for designing the converter circuit:

$$\frac{V_o}{V_{in}} = \frac{R_l * D * (1-D)}{2 * f_s * L * n}$$

Equation 1. Relation between output voltage and input Voltage

$$D = \frac{\emptyset}{T_s/_2}$$

Equation 2. Duty Ratio definition in phase difference and switching period

$$P_o = \frac{V_o * V_{in} * D * (1-D)}{2 * f_s * L * n}$$

Equation 3. Output power of converter

$$I_o = \frac{V_{in} * D * (1-D)}{2 * f_s * L * n}$$

Equation 4. Output current of converter

$$\Delta V_c = \left[\frac{D * (1-D)}{4 * C * f_s^3}\right] \left[\frac{V_{in}}{n * L}\right] \left[\left(4 f_s - D * (1-D)\right) - \frac{3 R_l * D * (1-D)}{8 * n^2 * L}\right]$$

Equation 5. Voltage ripple at output port

Where,

V_o = Output Voltage

V_{in}= Input Voltage

R_l= Resistive Load

D= Duty Ratio

f_s= Switching Frequency

L= Leakage Inductance Value

n= Transformer turns ratio (Secondary to Primary)

$\emptyset$= Phase shift in radian

T_s= Switching Time Period

P_o= Output Power

ΔV_c= Output Voltage Ripple

C= Filter Capacitance

Keeping the voltage ripples at 2%, and keeping in mind the resolution of the phase shift at the rated power for achieving better control over the plant, the elements are sized appropriately.

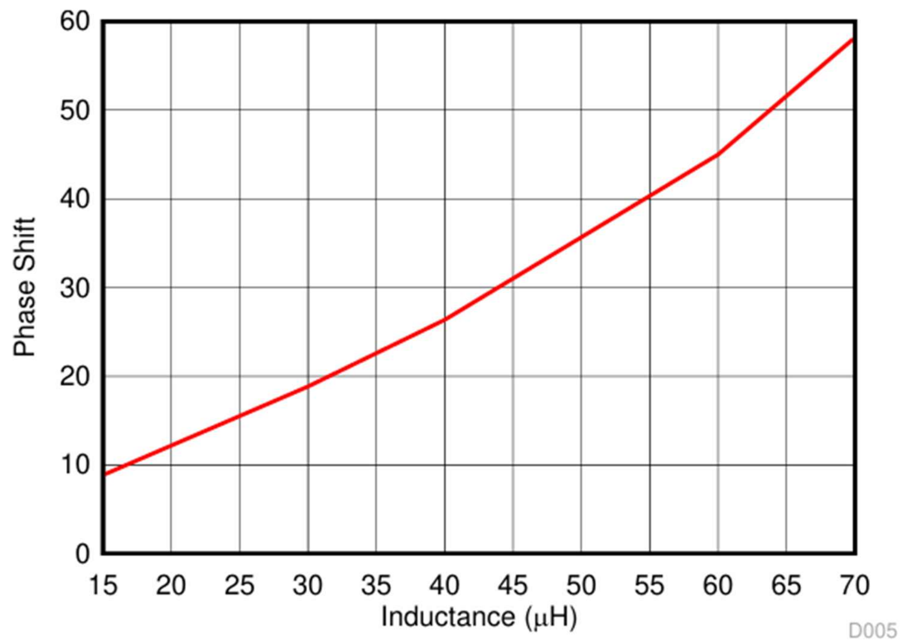


Figure 6- Phase Shift Resolution vs Inductance value for a Rated Power [10]

Figure 6 illustrates the Phase shift difference between the two bridges and the required inductance for the rated power of 500W. It is shown how to achieve more resolution and control over the phase shift, the size of the inductor is required to increase. The phase shift is stated in degrees in the above graph.

The controller proportional Kp and Ki values were chosen based on requirements of settling time of the switches(taken as 1 second) (bandwidth of the arms) and the inherent critically damped response of the system (Damping ratio = 1 (critical damping)) as mentioned earlier. This gives Kp= 8 and Ki=16. However, there was a bit more fine-tuning involved to include the aberrations of the plant model to finally reach an acceptable response. State space averaging of plant derived the following eqn.

$$\begin{bmatrix} V_c{}' \\ I_L{}' \end{bmatrix} = \begin{bmatrix} \frac{-1}{RC} & \frac{1-2D}{nC} \\ \frac{-(1-2D)}{nL} & 0 \end{bmatrix} \begin{bmatrix} V_c \\ I_L \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} [V_{in}]$$

Equation 6. Large signal Model of DAB Converter

$$i_{L'} = \frac{-(1-2D)}{nL}\ (V_0) + \frac{V_{in}}{L}$$

Equation 7. Inductor current from State Space Model

$$sI_L(s) = \left(\frac{nV_{in} - (1-2D)(V_0)}{nL}\right)$$

$$\frac{sI_L(s)}{m(S)} = \frac{1}{nL}$$

$$\frac{I_L(s)}{m(s)} = \frac{1}{s(nL)}$$

Equation 8. Simplified Plant Transfer Function

$$\frac{m - nV_{in}}{2V_0} + \frac{1}{2} = D$$

Equation 9. Value of D being used for Control of DAB Converter

3.1.1 Realistic Constraints

In the context of realistic constraints for the designed Dual Active Bridge (DAB) converter, one significant challenge is the availability of filter components. The specific filter sizes and inductance values required for the system's optimal performance might not be commercially available, making the procurement of such elements difficult and requiring potential custom fabrication. This limitation imposes added complexity and time constraints on the implementation of the design.

Furthermore, while simulations often assume instantaneous switching for simplicity, real-time operation introduces practical considerations such as dead time. This dead time, ideally set to approximately one-tenth of the half switching frequency, is necessary to allow the inductor current to settle to zero before the complementary arm of the H-bridge is activated. Without this provision, there is a risk of core saturation and excessive stress on the inductor, potentially leading to inefficiencies and damage to the components. These real-world factors must be taken into account to bridge the gap between theoretical designs and practical applications.

The design of the Dual Active Bridge (DAB) converter faces practical challenges when it comes to achieving precise phase shift resolution. This is crucial for optimal power flow but demands high-performance microprocessors and controllers capable of handling complex calculations in real-time. The computational burden of deducing the accurate phase shift increases with the desired resolution, which directly affects the converter's ability to modulate power efficiently. Such advanced control systems may not only elevate the overall cost of implementation but also require careful tuning and robust algorithms to ensure seamless operation under varying conditions. These factors highlight the intricacies involved in transitioning from theoretical design to practical realization of the DAB converter.

3.1.2 Alternatives and Trade-offs

In the controller design, the use of a simplified plant model introduced inaccuracies in the closed-loop performance of the system. These errors stemmed from the model's inability to capture the finer dynamics of the actual plant, leading to deviations from the desired control behavior. To address this limitation, a feedforward compensation technique was implemented as an alternative. By directly accounting for disturbances and anticipated system responses, the feedforward method enhanced precision and reduced reliance on the oversimplified plant model, thereby improving the overall performance of the control system.

Regarding trade-offs, employing a small leakage inductance to minimize the physical size of the converter posed challenges in achieving phase shift resolution, particularly at higher switching frequencies. The reduced inductance constrained the resolution, which is critical for precise power modulation and efficient operation. Additionally, the transformer designed for galvanic isolation introduced another trade-off due to its imperfect coupling. This resulted in unintended phase shifts that complicated power flow optimization, underscoring the balance between isolation requirements and power transfer efficiency. These compromises illustrate the intricate design decisions required to achieve a practical and robust system.

3.2 DESIGN SPECIFICATIONS

Parameters	Value
Input Voltage	400V
Output Voltage (Nominal)	60V
Output Power (Nominal)	500W
Switching Frequency	100KHz
Assumed Internal Resistance of Battery	1.2 ohms
Transformer Turn Ratio (1:n)	1 : 0.15
Inductor	35μH
Capacitor (Output, Input)	$1.35mF$, 10mF

Table 1- Design Specifications for Dual Active Bridge Converter

CHAPTER IV

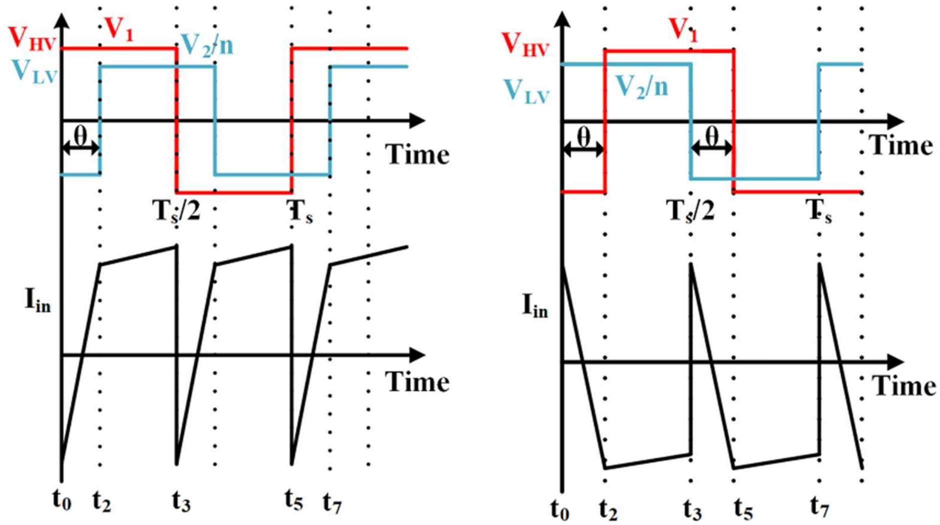
4. PROJECT DEMONSTRATION

4.1 INTRODUCTION

The Dual Active Bridge (DAB) converter operates by utilizing phase shift control between its primary and secondary H-bridge circuits to enable bidirectional power transfer. A key principle of its operation is the introduction of a phase lag between the voltage waveforms generated by the primary and secondary bridges. This phase lag results in a flow of power from the leading bridge to the lagging bridge, driven by the voltage difference across the transformer leakage inductance. The magnitude and direction of the phase shift determine the amount and direction of power flow, allowing the DAB converter to efficiently transfer energy between two voltage sources.

4.2 ANALYTICAL RESULTS

According to the theoretical analysis of the Dual Active Bridge (DAB) converter, when the second bridge lags the first bridge, the current through the leakage inductance exhibits a skewed quasi-trapezoidal waveform. This waveform arises due to the phase difference between the primary and secondary bridges, which induces a voltage difference across the leakage inductance, driving the current. Over one half-switching period, this quasi-trapezoidal current waveform becomes the repetitive pattern for the input port current, directly influencing the power transfer characteristics of the converter. The nature of this periodic waveform allows for mathematical representation, enabling the derivation of an expression for the average inductor current. By integrating the current waveform over a half-switching cycle and normalizing it with respect to the switching period, the equation for the average inductor current can be obtained. This expression serves as a crucial tool for analysing and designing the power flow and control strategies of the DAB converter.



(a) (b)

Figure 8-Input port Current at Forward (a) and Reverse (b) power flow conditions [10]

Figure 8 displays the variation in the input port current for varying conditions of power flow i.e. Forward and Reverse conditions. It can be seen that the average value of the current for the forward power flow is positive while negative for reverse power flow.

This information can be used to control the power flow in the converter. Analysing the waveform for the switching cycle and finding the average input current we get the equation below.

$$i_{inavg} = \frac{V_2}{2n\pi f_s L}\theta(1-\frac{\theta}{\pi})$$

Equation 10. Average Current through Input port [10]

4.3 SIMULATION RESULTS

Figure 7 displays the SIMULINK Model of the converter. Inbuilt blocks of the pulse generator and transport time delay and delay blocks were utilized with the function block of SIMULINK to create a plug-and-play model of the converter that is easy to simulate and understand. Required displays have been placed alongside the scopes for easy visibility of the outputs of the scope and real-time estimation during simulation.

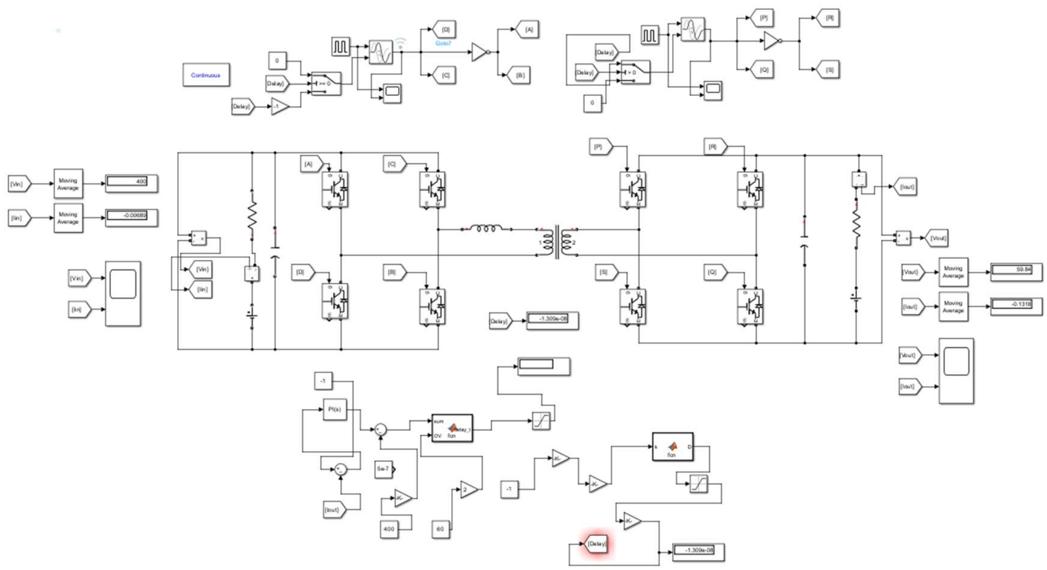


Figure 7- Simulink Flow of Converter with Control System

a) Open Loop Simulation

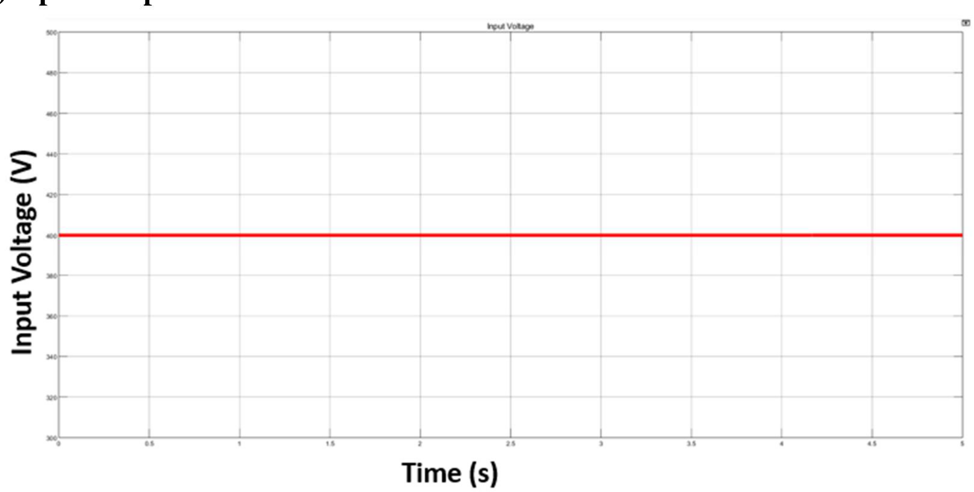


Figure 9 – Input Voltage of Dual active bridge Converter

Figure 9 displays the input voltage of the ideal DC Voltage source of 400V.

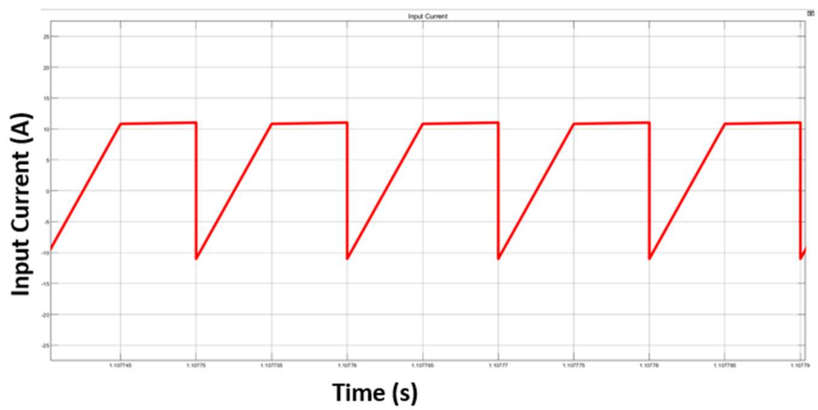


Figure 10- Input Current of Dual Active Bridge Converter

Figure 10 displays the Input port current. This matches the theoretical analysis we made in Figure 8.

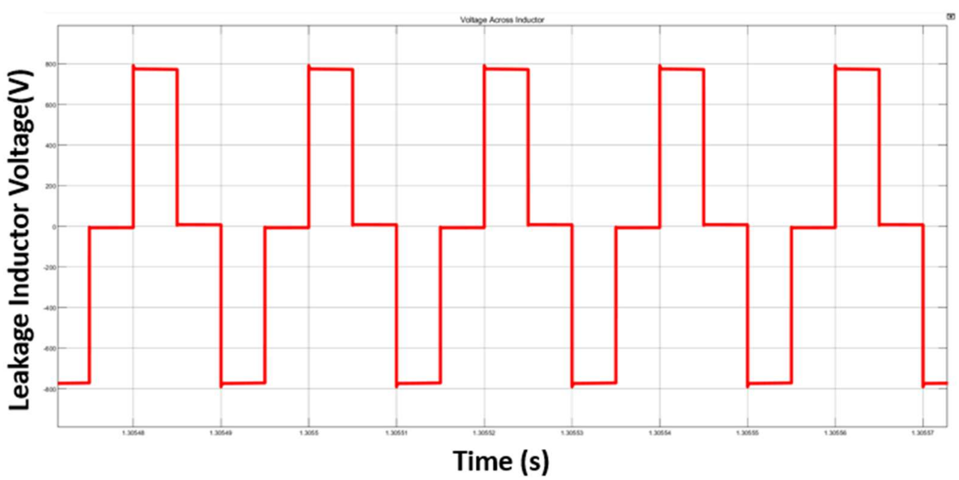


Figure 11- Voltage across the Leakage Inductor of Dual Active Bridge Converter

Figure 11 displays the Voltage across the leakage inductance at the critical condition $V_{in} = n * V_O$.

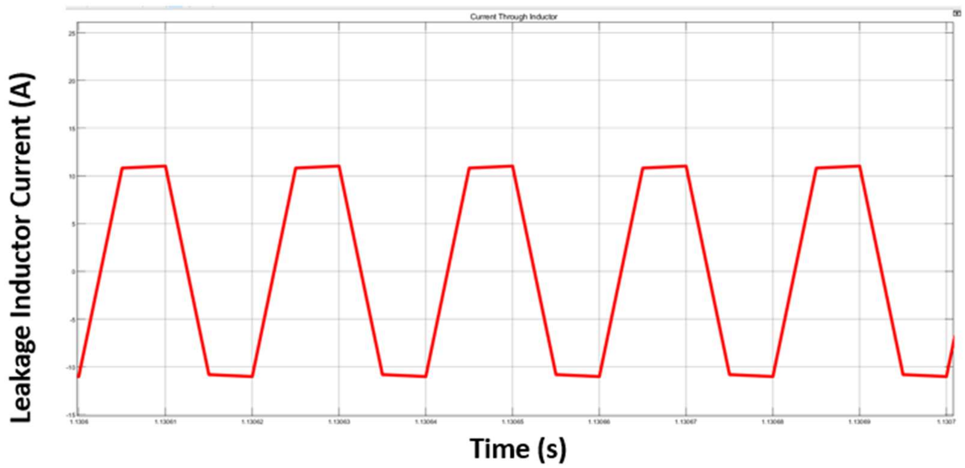


Figure 12- Current through Inductor of Dual Active Bridge Converter

Figure 12 displays the current through the leakage inductance. Each half of the switching period also depicts the input port current. Where the negative current of the inductor is passed through the freewheeling diode of the FET.

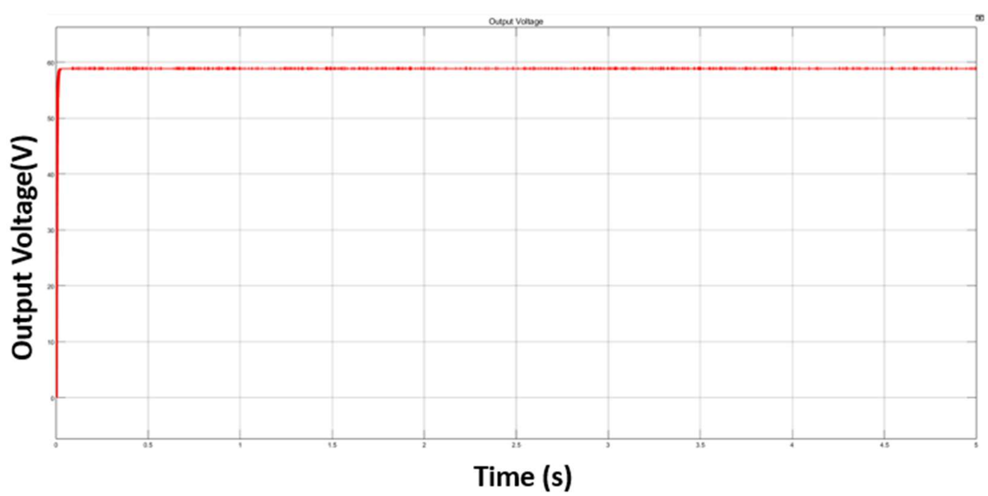


Figure 13- Output Voltage of Dual Active Bridge Converter

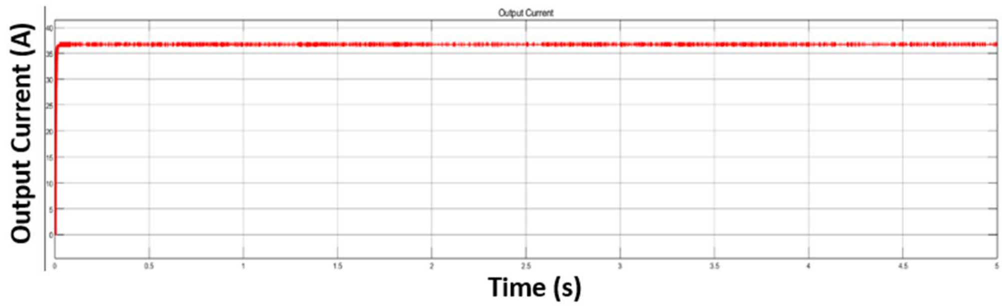


Figure 14- Output Current of Dual Active Bridge Converter

Figure 13 and 14 display the output current and voltage for a resistive load.

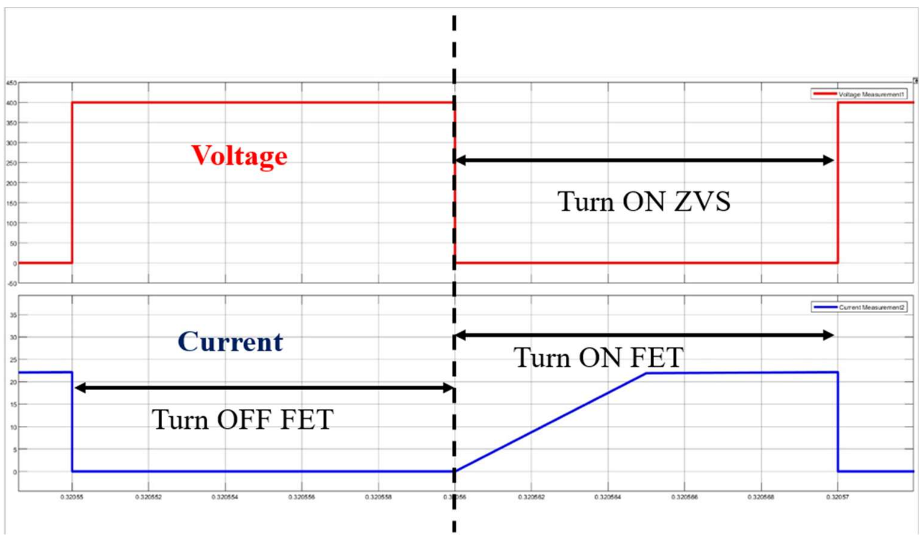


Figure 15- Soft Switching (ZVS) of Dual Active Bridge (DAB) FETs

Figure 15 displays the ZVS of the FETs of the DAB Converter. This inherent ZVS makes DAB a preferable choice in high-frequency power applications which is bolstered with the galvanic isolation provided by the transformer.

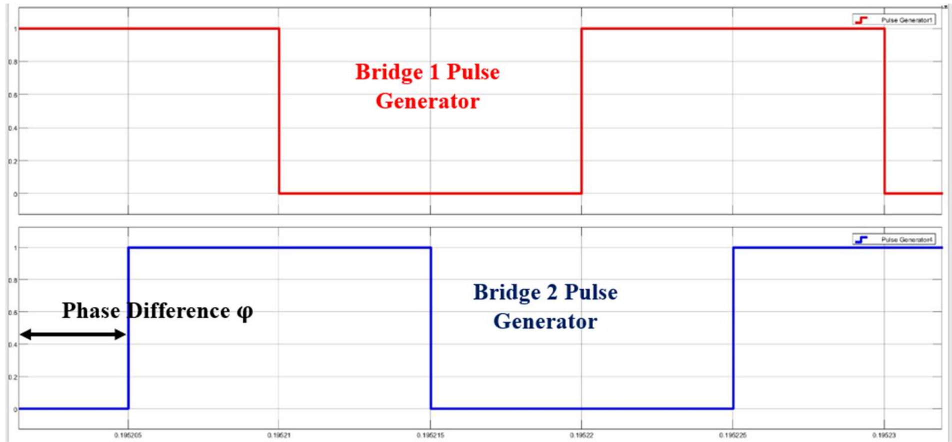


Figure 16-Pulse comparison for Bridge 1 and Bridge 2

Figure 16 displays the phase lag between the pulse generator bridges of the primary and secondary bridges of the DAB Converter. A positive lag (with lag leaning to the second bridge) depicts a forward power flow and a negative lag depicts a reverse power flow.

b) Closed Loop Simulation Results

- **4-Wheeler Battery (Source) to 2-Wheeler Battery (Load)**

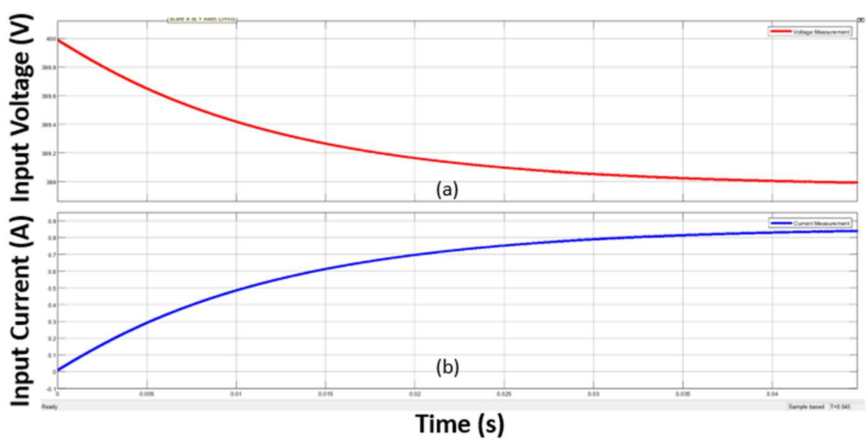


Figure 17-Voltage (a)and Current (b) Waveforms of Source

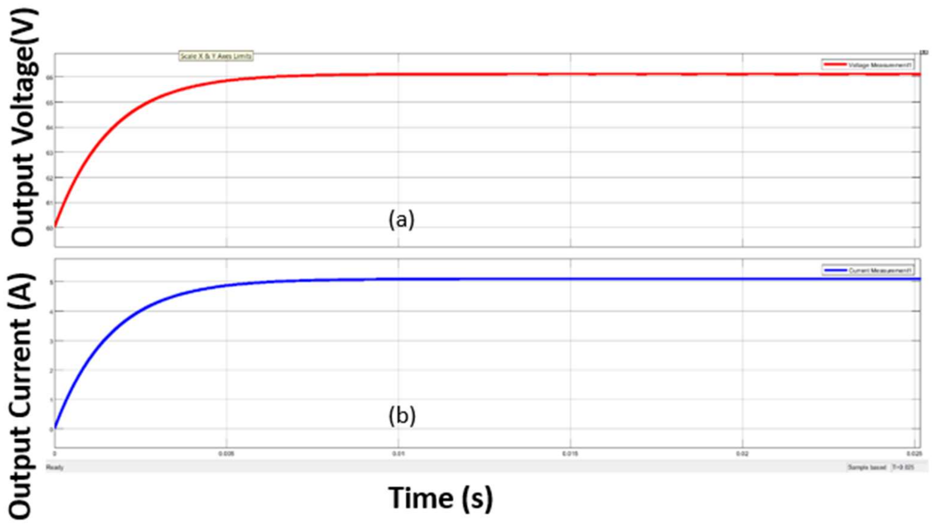


Figure 18- Voltage(a) and Current(b) Waveforms of Load

Figures 17 and 18 depict the forward power flow behavior in DAB converter, emphasizing the stabilization of electrical parameters at different ports. In Figure 17, the input voltage settles at **399 volts**, while the current stabilizes at **0.8 amps**, showcasing the system's steady operation at the input. Meanwhile, Figure 18 illustrates the output port dynamics, where the voltage stabilizes at **66 volts** and the current reaches **5 amps**, ensuring consistent power delivery to the load.

- **2-Wheeler Battery (Source) to 4-Wheeler Battery (Load)**

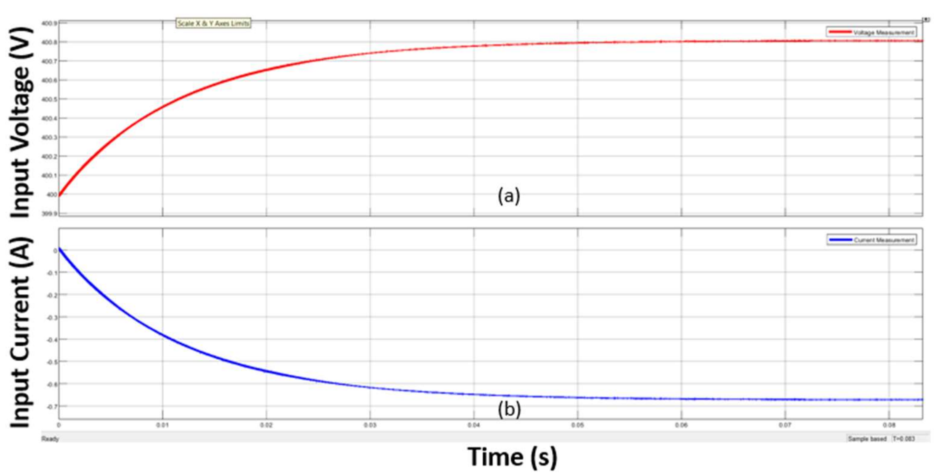


Figure 19-Voltage(a) and Current(b) Waveforms of Load

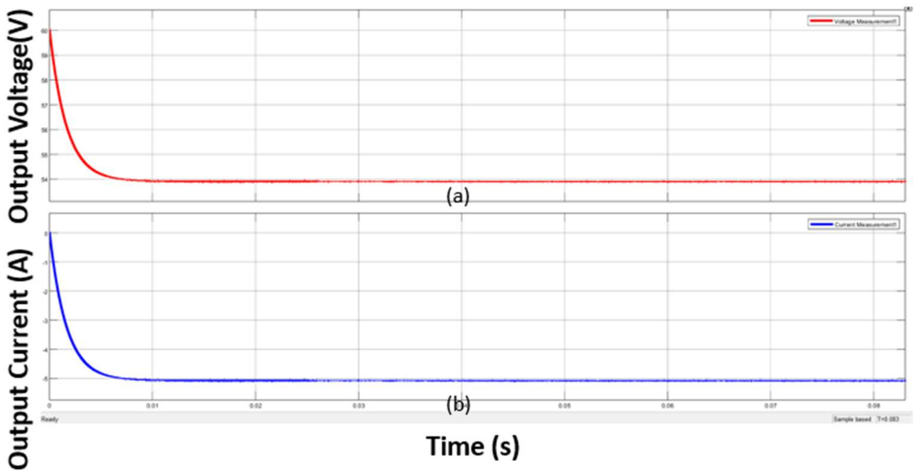


Figure 20-Voltage(a) and Current(b) Waveforms of Source

Figures 19 and 20 demonstrate the behavior of the dual active bridge (DAB) converter during reverse power flow, highlighting stabilized voltage and current values. In Figure 19, the input voltage settles at approximately **400.8 volts**, while the current stabilizes at **-0.7 amps**, indicating the direction of power flow at the input port. Similarly, Figure 20 illustrates the output port dynamics, where the voltage stabilizes at **54 volts**, and the current reaches **-5 amps**, reflecting consistent reverse power transfer to the output. Together, these figures showcase the DAB converter's efficiency in handling reverse power flow with stable performance across its input and output ports.

CHAPTER V

5. CONCLUSION

The DAB converter project successfully demonstrates a robust design capable of facilitating bidirectional power transfer between the 400V and 60V voltage ports. The converter utilizes the leakage inductance of the transformer as a key parameter for controlling power flow. This innovative approach ensures efficient energy transfer across the voltage ports while maintaining a compact design. The system is designed to regulate the energy exchange dynamically, meeting the functional requirements of both charging and discharging scenarios seamlessly. The converter achieves bidirectional power transfer using a negative reference current to generate the appropriate phase shift, ensuring precise control over power flow while safeguarding system stability.

Extensive simulation work has been conducted and validated using MATLAB Simulink to showcase the functionality and efficiency of the DAB converter. The simulation results highlight the converter's capability to achieve smooth and controlled bidirectional power transfer, adapting effectively to various operating conditions. Key metrics, including phase shift accuracy and power regulation, have been thoroughly tested and demonstrated, showcasing the practicality and reliability of the design. The project's success underscores the potential of leveraging advanced simulation tools in optimizing and verifying power electronics systems for real-world applications.

5.1 FURTHER SCOPE OF WORK

Replacing the ideal battery sources used in simulations with more realistic battery models from SIMSCAPE can provide a more accurate representation of battery dynamics. Such models incorporate detailed parameters, enabling the simulation of state of charge (soc), charging/discharging effects, and voltage drops under various conditions. This enhancement would allow the system to replicate real-world scenarios with improved fidelity, thereby offering valuable insights into battery behavior during energy transfer. By simulating actual SoC changes, researchers can better analyze battery performance, degradation, and efficiency under diverse operational loads, which is crucial for effective system design and optimization.

Additionally, predictive filtering techniques like Kalman filters could be introduced to estimate the battery's state of charge with higher precision, accounting for uncertainties and noise in real-world data. These filters can enable robust SoC determination, improving the reliability of energy management systems for electric vehicles. Furthermore, algorithms for driving range approximation can be integrated into the framework to estimate the driving range for two-wheeler and four-wheeler electric vehicle batteries. Such algorithms would leverage battery characteristics, real-time soc, and predicted energy consumption patterns to provide actionable insights to users. Implementing these tools can significantly enhance the practical applicability of the system, paving the way for improved user experience and system efficiency in EV applications.

5.2 SUMMARY

The Dual Active Bridge (DAB) converter project showcases a highly efficient design capable of bidirectional power transfer between 400V and 60V voltage ports. By leveraging the leakage inductance of the transformer, the system facilitates controlled power flow while maintaining compactness and reliability. This design enables efficient energy transfer for both charging and discharging applications, using a negative reference current to produce the appropriate phase shift for bidirectional operation. The converter has been rigorously analyzed and validated through MATLAB Simulink simulations, providing evidence of its practical viability and effectiveness in power modulation under varying conditions.

Through the simulations, the DAB converter demonstrated smooth operation and precise control, ensuring that both voltage regulation and dynamic power management were achieved successfully. The project emphasizes the adaptability of the design to real-world scenarios, offering insights into its application in battery charging systems for electric vehicles and other energy storage solutions. The results affirm the converter's potential for practical implementation while laying the foundation for further research and development to refine and enhance its capabilities.

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